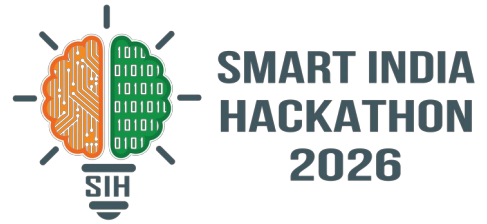


SMART INDIA HACKATHON 2026



Problem Statement ID: 26188 (SIH26188)

Problem Statement Title: AI-Based Fake Identity & Document Screening System

Ministry / Organization: Ministry of Home Affairs (MHA)

Theme: Blockchain & Cybersecurity / Smart Border Security

PS Category: Software

Idea Title: Checkpoint AI: Automated Identity & Document Screening System

Team ID: [Registered Team ID on SIH Portal]

Team Name: TEAM HELIX

Institute / College: Bhaskaracharya College of Applied Sciences, University of Delhi



IDEA TITLE

- ❖ ● **Core Challenges in Current Border Surveillance**
 - **Manual Bottlenecks:** High passenger volume causes 3–5 min clearance delays per traveler, overwhelming immigration counters.
 - **Undetected Forgeries:** Digital alterations, modified dates of birth, and forged visa stamps easily bypass routine human visual inspection.
 - **Identity Impersonation:** Validating likeness between fraudulent document portraits and stressed live travelers is error-prone under pressure.
 - **Missing Digital Trail:** Fragmented inspection logs prevent rapid background checks and inter-agency intelligence coordination.

THE CURRENT SYSTEM (WHY IT FAILS)	CHECKPOINT AI SOLUTION (HOW WE FIX IT)
[REACTIVE CHECKS] Manual screening performed only at physical counters during transit.	[PROACTIVE PRE-ARRIVAL] SmartVisa portal scanning processes credentials before arrival.
[SUBJECTIVE INSPECTION] Officers rely on visual inspection; subtle digital edits evade detection.	[AI-DRIVEN FORENSICS] Error Level Analysis (ELA) exposes pixel-level Photoshop manipulations & splicing.
[MANUAL ENTRY ERRORS] Slow keying of traveler details prone to fatigue and human typos.	[AUTOMATED PIPELINE] End-to-end processing pipeline generates an explainable 0–100 Risk Score in seconds.

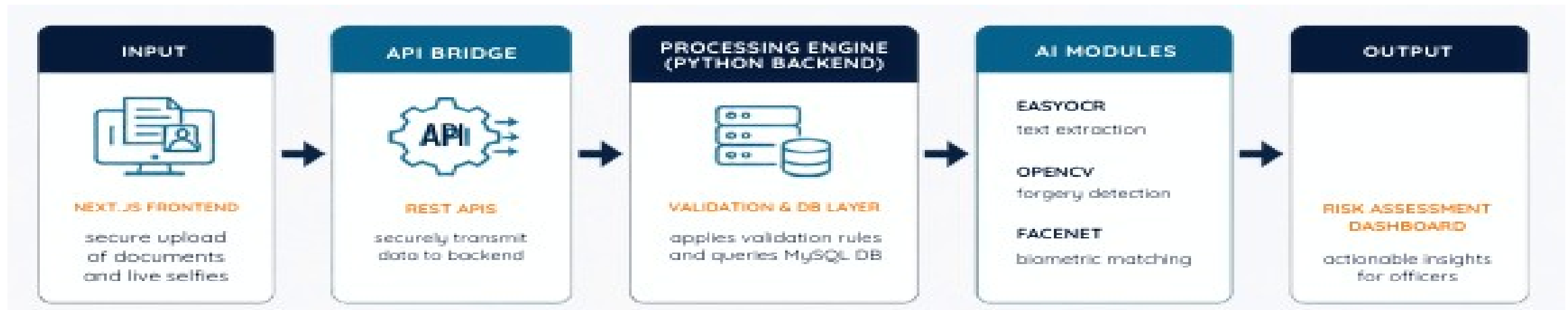
Innovation & Uniqueness: Multi-layered fusion of dual-stream OCR (ICAO 7-3-1 check-digits vs VIZ), compression artifact heatmaps (ELA), and 1:1 biometric facial liveness into an intuitive, zero-training risk assessment console.

TECHNICAL APPROACH

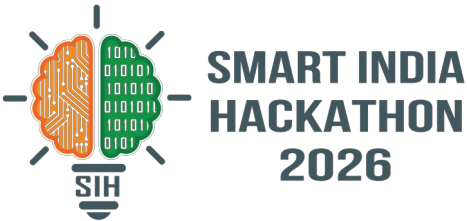
- **AI Forensic Modules & Deep-Dive Capabilities**

- **EasyOCR & Dual Extraction:** Automated text retrieval from Machine Readable Zone (MRZ Lines 1 & 2) and Visual Inspection Zone (VIZ); performs cryptographic 7-3-1 modulus-10 check-digit cross-validation.
- **Error Level Analysis (ELA):** Analyzes JPEG recompression discrepancies across error grids to generate visual heatmaps exposing Photoshop edits, clone stamping, and spliced text fields.
- **FaceNet & 1:1 Biometric Match:** Extracts 512-D deep facial embeddings from document portrait and live passenger selfie; computes cosine similarity (96% benchmark) with passive anti-spoof liveness.
- **ICAO Doc 9303 Compliance:** Strict algorithmic verification against international MRTD standards (TD1, TD2, TD3) with instantaneous document classification and gatekeeping.

- **5-Stage End-to-End System Workflow**



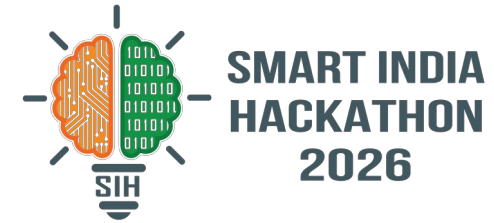
FEASIBILITY AND VIABILITY



- **Feasibility & Viability Analysis**
 - **Technical Feasibility:** Built on open, battle-tested stacks (Next.js, Python FastAPI, OpenCV, EasyOCR, FaceNet). Delivers low-latency (<2s) inference on standard commodity checkpoint PCs without expensive GPU clusters.
 - **Operational Viability:** Seamlessly retrofits into existing immigration booths using standard flatbed scanners and webcams. Zero specialized training needed; intuitive green/amber/red indicators.
 - **Economic Viability:** 100% software-based solution eliminating millions in capital expenditure for proprietary spectral hardware scanners at high-traffic land borders.
- **Challenges & Risk Mitigation Strategies**
 - **Scan Glare & Faded Paper:** Automated CLAHE (Contrast Limited Adaptive Histogram Equalization) and unsharp masking image enhancement prior to OCR processing.
 - **Remote Border Connectivity:** Hybrid Edge-First architecture: local offline verification on checkpoint PCs with automatic background synchronization to central databases once reconnected.
 - **Biometric Spoofing (Print/Screen):** Multi-frame temporal liveness analysis (micro-blink and landmark depth heuristics) blocks printed photographs and mobile screen replays.

ID	DOMAIN / TRACK	RESPONSIBILITIES & DELIVERABLES
01	FRONTEND & ARCHITECTURE	Next.js officer dashboard, responsive UI, and end-to-end system flow management
02	AI LOGIC & COMPUTER VISION	Model training, pixel-level tampering detection integration, and ELA heatmap generation
03	BACKEND & DATABASE	Python REST APIs, rule validation engine, and secure verification data handling
04	RESEARCH & COMPLIANCE	Ensuring alignment with border protocols, BPR&D guidelines, and ICAO 9303 standards
05	QA & TESTING	Prototype stress-testing with spoofed datasets aiming for zero false positives

IMPACT AND BENEFITS



Tagline: Smarter verification. Stronger security. Scalable for the future.

- **Strategic National Security & Socio-Economic Benefits**

- **National Sovereignty & Defense:** Directly prevents illegal cross-border migration, human trafficking, and fake passport syndicates under the Ministry of Home Affairs (MHA).
- **Auditable Forensic Dossiers:** Automatically generates 1-click legal PDF investigation reports with highlighted tamper heatmaps and cryptographic timestamps for intelligence coordination.
- **Economic & Resource Efficiency:** Drastically reduces manpower expenditure while increasing passenger clearance capacity across India's land, sea, and air borders.

10 SECONDS

VERIFICATION TIME

Reduced from ~5 minutes to under 10 seconds per traveler, eliminating transit terminal bottlenecks.

ENHANCED SECURITY

OBJECTIVE DECISIONS

Standardized, data-driven AI risk scores replace subjective, fatigue-prone manual checks.

SCALABLE INTEGRATION

SMART INFRASTRUCTURE

Compatible with CCTV infrastructure, immigration databases, and automated e-Gates via QR codes.

Checkpoint AI delivers measurable impact at borders — **we are ready to innovate and implement with confidence.**

- **International Standards & Specifications**

- **ICAO Doc 9303 (Parts 3, 4, 7):** International Civil Aviation Organization specifications for Machine Readable Travel Documents (MRTDs); standardized TD1/TD2/TD3 zones and 7-3-1 weight modulus-10 check-digit algorithms.
- **ISO/IEC 19794-5 Biometric Data Interchange:** Information Technology — Biometric data interchange formats — Part 5: Face image data quality, inter-pupillary distance, landmark standards, and frontal pose compliance.

- **Digital Forensics & Computer Vision Research Work**

- **Error Level Analysis (ELA):** Krawetz, Neal (2007). 'A Picture's Worth... Digital Image Analysis and Error Level Analysis (ELA)', Hacker Factor Solutions — Foundations of JPEG recompression forensics.
- **Deep Facial Embeddings (ArcFace & FaceNet):** Schroff, Kalenichenko, Philbin (2015). 'FaceNet: A Unified Embedding for Face Recognition' & Deng et al. (CVPR 2019) ArcFace additive angular margin loss for extreme verification accuracy.
- **Document Tampering Localization:** Bappy, Roy-Chowdhury, et al. (IEEE TIFS 2019). 'Exploiting Spatial Structure for Localizing Manipulated Image Regions' — Multi-scale patch analysis for digital splicing.

- **National Guidelines & Policy Frameworks**

- **Ministry of Home Affairs (MHA):** Integrated Border Management System (IBMS) Guidelines for Smart Border Checkpoints.
- **Bureau of Police Research & Development (BPR&D):** Forensic Questioned Document Examination Manual and Passport Authentication Directives.